



From Contamination to Compliance: The Role of Sartorius Lab Water System in PFAS Detection

Reduction of PFAS Contaminants Using Arium® Comfort I UV

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Per- and Polyfluoroalkyl Substances (PFAS), a family of over 4,700 chemical molecules is known for non-stick and waterproof properties. These versatile compounds find their application in diverse industries, including cosmetics, textiles, and automobiles. Resistance to degradation, they persist in water, soil and atmosphere, contributing to environmental concern.

This application note uses two complementary analytical techniques: liquid chromatography – mass spectrometry

and combustion ion chromatography to discern how the Arium® Comfort I UV reduces PFAS contaminants concentration below detection limits to up to single-digit ppt or sub-ppt-level (parts per trillion, dependent on the compound) in water. The results show that Arium® Comfort I UV is highly efficient in the reduction of PFAS in water with both the standard and PTFE-free setup.

Keywords or phrases: Per- and Polyfluoroalkyl Substances (PFAS), PFAS reduction, Ultrapure water, ASTM Type I water, Arium®, LCMS, CIC TOF

Introduction

PFAS are a group of synthetic chemicals widely used in industrial and consumer products, such as non-stick cookware, waterproof clothing, and firefighting

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foam, due to their unique properties. However, their persistence in the environment and potential adverse health effects have raised concerns about their presence in water sources, including drinking water. Human exposure to PFAS can occur through ingestion of contaminated food and water, inhalation of dust and air, and skin contact with products containing PFAS.

Once released into the environment, PFAS can persist for a long time and can leach into soil and water,

concentration below detection limits to up to single-digit ppt or sub-ppt-level (parts per trillion, dependent on the compound) in water.

Ultrapure Water Purification

System Overview

The Arium® Comfort I UV is a comprehensive water treatment solution designed to produce ultrapure water from in- house potable water for various applications,

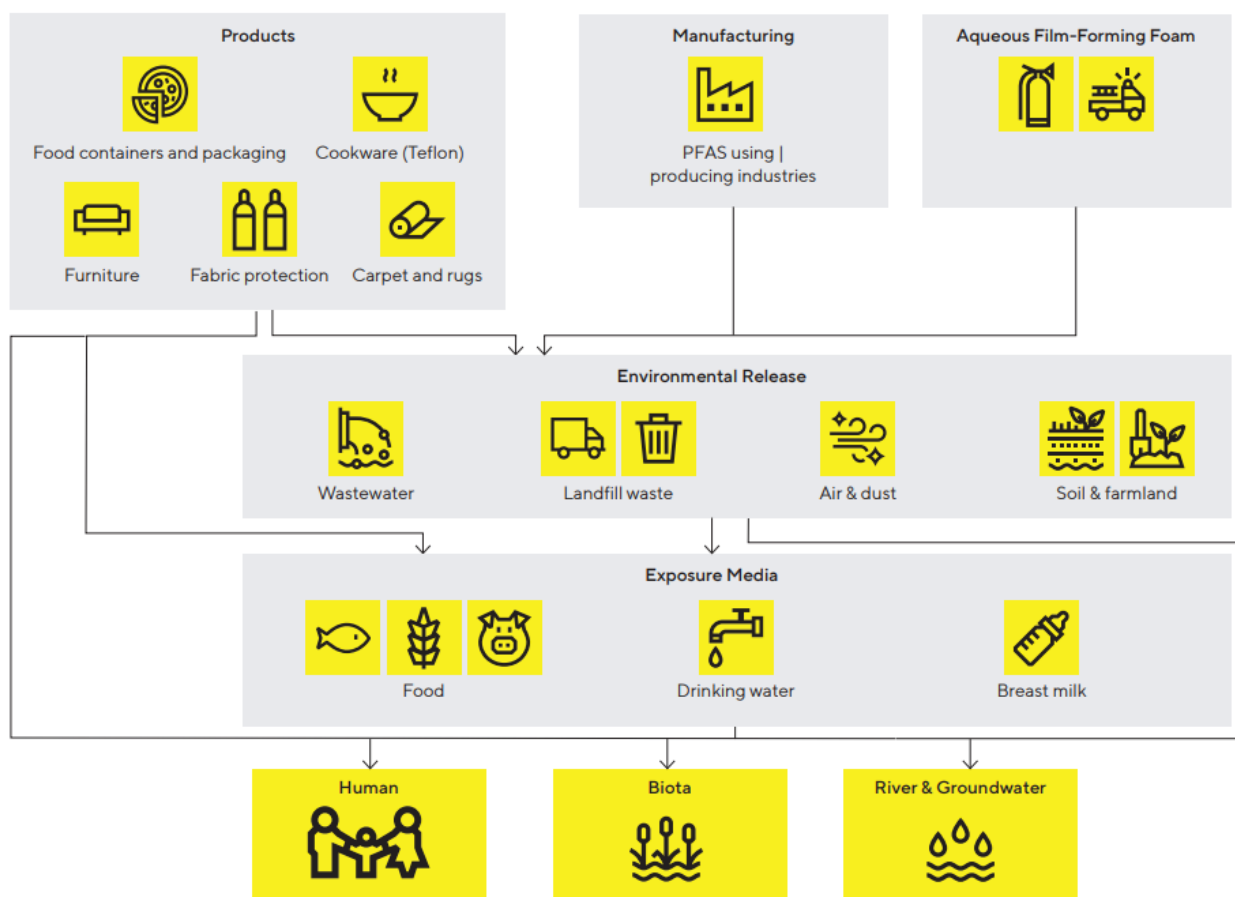


Figure 1: Main PFAS Exposure Pathways for Humans and Biota²

potentially contaminating drinking water sources and harming wildlife (Figure 1).^{1,2}

This application note discusses how the Arium® Comfort I UV effectively reduces PFAS contaminants

ensuring the reduction of a wide range of contaminants, including traces of metal ions, organic contamination and PFAS. The system employs a multi-stage approach combining activated carbon, reverse osmosis technology, and resin ion exchange to achieve efficient PFAS removal.

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PFAS Removal Process

PFAS Removal Process³

Our Arium® Comfort I UV employs a multi-stage process to eliminate PFAS contaminants from potable water, ensuring the highest level of water purity.

Reverse Osmosis (RO) Module

The process begins with the water passing through the RO module. The cornerstone of our PFAS removal strategy lies in the implementation of advanced RO technology.^{4, 5} RO has proven to be remarkably efficient in removing PFAS compounds, boasting a removal rate of up to 99% of PFOS from the feedwater.⁶ This includes the drastic reduction of both long-chain and short-chain PFAS molecules (C₄ - C₁₄), demonstrating the technology's comprehensive efficacy.

Activated Carbon Filtration

Following RO filtration, the water enters the activated carbon filtration stage. Activated carbon, renowned for its exceptional adsorption properties, further targets PFAS molecules.

This step enhances the purification process by capturing remaining PFAS compounds and other organic impurities.^{7, 8}

Resin Ion Exchange

The final stage involves resin ion exchange, a process that further polishes the water by exchanging undesirable ions, including remaining traces of PFAS, with ions on the resin.

This step ensures that even the minute concentrations of PFAS that may have passed through earlier stages are effectively removed.⁴

Achieving Below Detection Limit Removal

The combination of activated carbon, RO treatment, and

resin ion exchange in the Arium® Comfort I UV enables the reduction of PFAS concentrations to below detection limits in tap water. The system ensures the production of high-quality water that is verified to reduce the risk of PFAS contamination from the original water source, meeting the stringent purity requirements of various applications.

Treating Highly Contaminated PFAS Solutions

The effectiveness of the Arium® Comfort I UV is not limited to potable water. The system has been tested and proven to treat highly contaminated solutions containing up to ~4 ppb (parts per billion) of various PFAS compounds. This capability highlights the system's robustness and efficiency in addressing challenging contamination scenarios.

Materials

- › Arium® Comfort I UV
- › Arium® Bag Tank
- › LCMS System (Shimadzu 8060 LCMS and PromoChromSPE03)
- › CIC-System (Metrohm Custom Halogen Analyzer)
- › Extracting tools and equipment
- › Arium® Comfort I UV

Sartorius offers the compact, environmentally friendly, reliable, and easy-to-use Arium®

Comfort I for producing ASTM Type 1 ultrapure water and Type 3 pure water combined in a single system. The system contains state-of-the-art reverse osmosis technology and a unique cartridge specifically to produce the highest ultrapure water quality.

Compared to conventional water systems, the Arium® Comfort I optimizes water consumption using the integrated iJust control unit. This unique touch display with intuitive menu navigation ensures the utmost ease of use.

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With the optionally integrated TOC monitor, its compact design, and the SD card slot, the Arium® Comfort I is the ideal choice for demanding laboratory applications.

Arium® Bagtank

The pure water is stored in the innovative enclosed Arium® Bagtank system. This system protects the prepared pure water against secondary contamination. The Sartorius Bagtank system enables consistent water quality over a prolonged period, thereby ensuring permanent, reproducible results. Unlike conventional water reservoirs, the Arium® Bag offers a high level of user safety and time savings, as there is no need for a complicated cleaning procedure with chemicals.

LCMS Condition

The instrument is modified to reduce PFAS contamination from the instrument components and LC solvents including a delay column. The chromatographic separation is performed with a BEH C18 column (2.1 × 100 mm, 2.1 µm) using a gradient (Solvent A and B follows EPA draft method 1633).

MS source conditions are adjusted to allow optimized performance for all 40 PFAS compounds.

Storage

Samples were stored in 4 °C environment prior to the extraction.

Extraction

Samples were taken out of the refrigerated environment and allow to return to room temperature. Method blanks and lab control samples were created according to the lab procedure. All samples, method blank, and lab control samples contain the same amount of spike of surrogate compounds (extracted internal standards). The samples were mixed by inverting the bottle gently backing forth for approximately one minute. The samples were loaded on PromoChrom SPE03 automatic solid phase extraction system. The 1633 method solid

phase extraction method was used for these samples.

CIC-Condition

Combustion Ion Chromatography (CIC) data was obtained using a customized halogen analyser from Metrohm for the analysis of Total Organofluorine (TOF) data per sample, allowing for a direct comparison with LCMS data. Briefly, a known amount of sample is combusted in a greater than 1,000 °C oven and the gas phase containing total ions is condensed in an absorber solution with subsequent analysis by ion chromatography. This result gives the total fluorine (TF) value of the sample. An Asupp7 column is used in standard conditions (EPA 300.0) to separate and quantify the components which are detected by conductivity.

Simultaneously, the same sample is sent directly to a different IC to analyse for inorganic fluorine (IF, as free fluoride). This result gives the inorganic fluoride value. TOF may now be calculated by $TF - IF$. The analysis is non-speciated, focusing on the collective impact of PFAS, and offers a streamlined approach with significantly reduced sample preparation. This comprehensive TOF perspective provides prompt insights into unknowns, potentially negating the need for subsequent advanced testing. In this analytical setup, the Custom Halogen Analyzer has been optimized after receipt by Metrohm and features a reporting limit of 7.5 ppb.

Result

LCMS Results

The method of detection limit (MDL) and the method reporting limits (MRL) for LCMS are shown in Table 1. The last complete validation of the LCMS was performed in Q2 2023, and the analysis were performed in Q2 and Q3 2023. Any compounds with T in the analyte indicate that both branched and linear isomers are in quantitation consideration.

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Analyte	MDL (ppt)	MRL (ppt)
PFBA	2.1	5
PFPeA	0.399	1
PFHxA	0.375	1
PFHpA	0.419	1
PFOA	0.507	2
PFNA	0.547	2
PFDA	0.274	1
PFUnA	0.399	1
PFDoA	0.554	2
PFTTrDA	0.759	2
PFTTeDA	0.395	1
T-NMeFOSAA	0.212	1
T-NEtFOSAA	0.514	2
PFBS	0.503	2
PFPeS	0.489	2
T-PFHxS	0.427	1
PFHpS	0.5	2
T-PFOS	0.351	1
PFNS	0.395	1
PFDS	0.373	1
PFDoS	0.748	2
4:2 FTS	1.389	5
6:2 FTS	2.215	5
8:2 FTS	2.55	10
PFOSA	0.445	1

Analyte	MDL (ppt)	MRL (ppt)
HFPO-DA	0.628	2
ADONA	0.973	2
9Cl-PF3ONS	0.662	2
11Cl-PF3OUdS	1.958	5
N-MeFOSA	0.612	2
N-EtFOSA	0.672	2
N-MeFOSE	10.668	30
N-EtFOSE	6.733	20
3:3 FTCA	0.412	1
5:3 FTCA	3.099	10
7:3 FTCA	3.483	10
PFMPA	0.314	1
PFMBA	0.35	1
NFDHA	0.907	2
PFEESA	0.308	1

Table 1: MDL and MRL Values of the PFAS

Experiment I

Samples of water mimicking a normal lab setting by connecting the system to the tap water. Two different Arium® Systems were chosen for this experiment. The first Arium® Comfort UV I system has the standard setup, which is equipped with virgin PE and PTFE tubing (Setup I).

The second Arium® Comfort I UV system is only equipped with PE tubing (Setup II). The samples were collected at (Figure 2):

- Feed water
- Post RO-treatment and before the water was stored in Arium® Bagtank
- At point of use of the system (without any final filter)

Results I

The Arium® Comfort I UV systems were installed in Woodbury, MN, a location with a historical presence of PFAS in tap water.

The analysis conducted three times and detailed in Table 2 with results reported in ppt, reveals that the PFAS content in the feed water is consistently reduced to levels below the limit of detection after undergoing RO-treatment.

Furthermore, there is no discernible PFAS leakage during the polishing step of the system.

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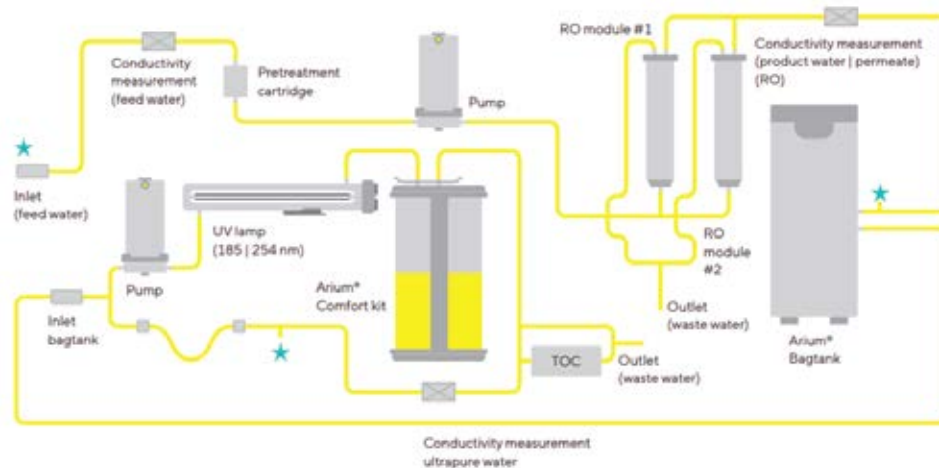


Figure 2: Flowchart of Arium® Comfort I UV. Blue Stars Mark the Point of Sampling

	Feed Water (ppt)		Post RO (ppt)		Point of Use (ppt)	
	Average	Standard Deviation	Setup I	Setup II	Setup I	Setup II
PFBA	233.715	1.778	n.d.	n.d.	n.d.	n.d.
PFPeA	5.275	0.102	n.d.	n.d.	n.d.	n.d.
PFHxA	1.796	0.026	n.d.	n.d.	n.d.	n.d.
PFOA	1.982	0.087	n.d.	n.d.	n.d.	n.d.
PFBS	1.374	0.055	n.d.	n.d.	n.d.	n.d.
T-PFHxS	1.451	0.277	n.d.	n.d.	n.d.	n.d.
T-PFOS	0.926	0.105	n.d.	n.d.	n.d.	n.d.
PFHpA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFNA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFDA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFUnA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFDoA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFTtDA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFTeDA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
T-NMeFOSAA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
T-NEtFOSAA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFPeS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFHpS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFNS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFDS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFDoS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
4:2 FTS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
6:2 FTS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
8:2 FTS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFOSA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
HFPO-DA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
ADONA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
9Cl-PF3ONS	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
11Cl-PF3OUDs	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
N-MeFOSA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
N-EtFOSA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
N-MeFOSE	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
N-EtFOSE	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
3:3 FTCA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
5:3 FTCA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
7:3 FTCA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFMPA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFMBA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
NFDHA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
PFEESA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.

Table 2: PFAS Results of the Experiment I. The Table Shows the Average of Three Analysis Content of the Respective PFAS

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Experiment II

Samples for the spiked the 50 L bag filled with reagent water with additional PFAS to test the system's capability were collected at (Figure 3):

- › The reagent water, before it was spiked with PFAS
- › Post 50 L bag for the spiked PFAS concentration
- › Post RO-treatment and before the water was stored in Arium® Bagtank
- › At the outlet of the system (without any final filter)

The first Arium® Comfort UV I system has the standard setup, which is equipped with virgin PE and PTFE tubing (Setup I).

The second Arium® Comfort I UV system is only equipped with PE tubing.

Results II

All reported results in Table 3 are in ppt. The PFAS concentration before spiking was below the limit of detection, consequently, is not presented in the table below. Although one 50 L water bag was intentionally spiked with PFAS, the experiments were conducted a week apart, resulting in an intriguing disparity in the initial concentrations. This variance could be attributed to the practical challenges of effectively mixing PFAS with a 50 L water bag, given the surfactant nature of PFAS, potentially resulting in a heterogeneous distribution within the bag. Furthermore, the instability of certain PFAS compounds at room temperature could contribute to the observed variations. These nuances underscore the importance of careful experimental design and consideration of PFAS properties in ensuring accurate and reliable results.

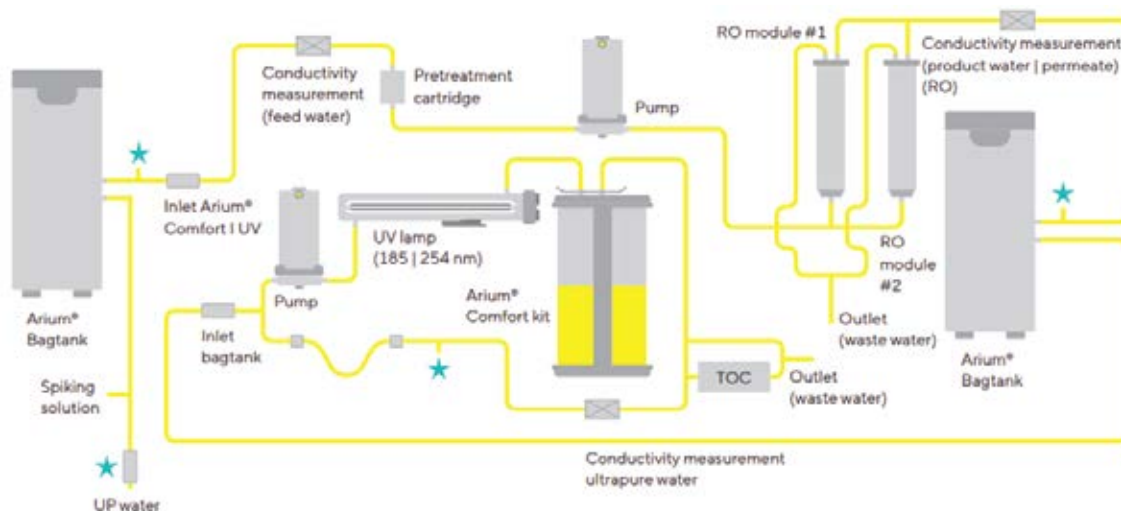


Figure 3: Modified Flowchart for the Spiked Experiment Utilizing Arium® Comfort I UV. Blue Stars Mark the Point of Sampling

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	Setup I			Setup II		
	Post 50 L bag (ppt)	Post RO (ppt)	Point of Use (ppt)	Post 50 L bag (ppt)	Post RO (ppt)	Point of Use (ppt)
PFBA	1166.33	n.d.	n.d.	4587.67	n.d.	n.d.
PFPeA	1090	n.d.	n.d.	4149	n.d.	n.d.
PFHxA	881.43	n.d.	n.d.	4482	n.d.	n.d.
PFHpA	17.63	n.d.	n.d.	106.83	n.d.	n.d.
PFOA	502.17	n.d.	n.d.	2017	n.d.	n.d.
PFNA	14.4	n.d.	n.d.	89.5	n.d.	n.d.
PFDA	15.1	n.d.	n.d.	94.77	n.d.	n.d.
PFUnA	11.67	n.d.	n.d.	94	n.d.	n.d.
PFDoA	7.33	n.d.	n.d.	68	n.d.	n.d.
PFTrDA	4.73	n.d.	n.d.	44.07	n.d.	n.d.
PFTeDA	2.87	n.d.	n.d.	21.27	n.d.	n.d.
T-NMeFOSAA	8.7	n.d.	n.d.	60.37	n.d.	n.d.
T-NEtFOSAA	6.3	n.d.	n.d.	37.53	n.d.	n.d.
PFBS	1387.33	n.d.	n.d.	8196	n.d.	n.d.
PFPeS	26.5	n.d.	n.d.	293.93	n.d.	n.d.
T-PFHxS	638.07	n.d.	n.d.	4103.33	n.d.	n.d.
PFHpS	47.5	n.d.	n.d.	389.47	n.d.	n.d.
T-PFOS	636.83	n.d.	n.d.	4271	n.d.	n.d.
PFNS	17.57	n.d.	n.d.	200.47	n.d.	n.d.
PFDS	11.33	n.d.	n.d.	181.5	n.d.	n.d.
PFDoS	4.3	n.d.	n.d.	49.37	n.d.	n.d.
4:2 FTS	62.07	n.d.	n.d.	342.57	n.d.	n.d.
6:2 FTS	54.33	n.d.	n.d.	280.57	n.d.	n.d.
8:2 FTS	52.37	n.d.	n.d.	303.23	n.d.	n.d.
PFOSA	1.87	n.d.	n.d.	n.d.	n.d.	n.d.
HFPO-DA	12.83	n.d.	n.d.	79.6	n.d.	n.d.
ADONA	12.57	n.d.	n.d.	82.2	n.d.	n.d.
9CI-PF3ONS	10.87	n.d.	n.d.	74.97	n.d.	n.d.
11CI-PF3OUdS	3.9	n.d.	n.d.	94.6	n.d.	n.d.
N-MeFOSA	0.73	n.d.	n.d.	n.d.	n.d.	n.d.
N-EtFOSA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
N-MeFOSE	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
N-EtFOSE	37.07	n.d.	n.d.	522.7	n.d.	n.d.
3:3 FTCA	343.23	n.d.	n.d.	4651.67	n.d.	n.d.
5:3 FTCA	131.3	n.d.	n.d.	1218.6	n.d.	n.d.
7:3 FTCA	24.63	n.d.	n.d.	322.03	n.d.	n.d.
PFMPA	21.4	n.d.	n.d.	260.47	n.d.	n.d.
PFMBA	39.1	n.d.	n.d.	523.3	n.d.	n.d.
NFDHA	35.53	n.d.	n.d.	494.13	n.d.	n.d.
PFEESA	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.

Table 3: Post 50 L Bag, Post RO Treatment and at the Point of Use Concentrations of the PFAS in the Spiked Experiment

Notable differences in initial concentrations between the two setups distinctly influence the concentrations observed at both the post RO sampling point and the point of use sampling point. Commencing at concentrations in part ppb, the analytes swiftly reach non-detect levels at the point of use. The Arium® Comfort I UV system, producing reagent water, further enhances its significance, providing water of the requisite purity for testing labs engaged in PFAS applications.

TOF Results

The determination of Total Organic Fluorine (TOF) involves the subtraction of Inorganic Fluorine (IF) from the total, resulting in TOF. To calculate TOF, the inputs required are TF and IF ($TOF = TF - IF$). The measurement of free fluoride in the sample typically represents IF (Figure 4).

The process of obtaining TF involves combusting the sample at temperatures exceeding 1,000 °C to break carbon bonds and form ions. The gas containing the ions

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of interest is then condensed into an absorption solution and analysed by Ion Chromatography (IC). Determining IF is achieved by running the sample through IC for free fluoride. The subtraction of these values yields the TOF of the sample.

It is crucial to consider various factors, including matrix effects, contaminant ions, and the presence of dissolved or suspended solids and metals. For solids, a different approach is necessary, requiring the extraction of a solid for IF determination even though TF can be directly determined.

Adhering to the same experimental design and utilizing identical sample points as outlined in the LCMS schema, the chosen methodology involves employing CIC to analyse TOF data on a per-sample basis, allowing for a direct comparison with LCMS data. This non-specified analysis is specifically focused on evaluating the combined impact of PFAS. A comprehensive examination of TOF provides swift insights into unknown elements, offering a rapid understanding.

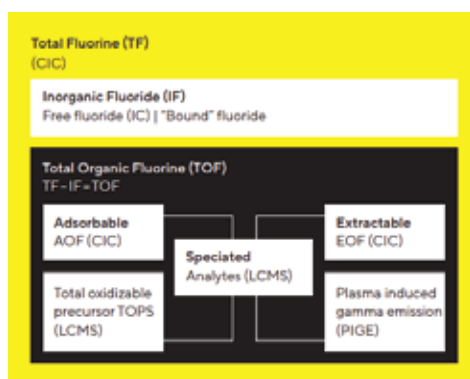


Figure 4: Scheme of a Fluorine Mass Balance Approach Applying Organically Bound Fluorine Sum Parameters

Sample ID		Setup I	Setup II
Experiment I	Tap Water (ppb)	<7.5	<7.5
	Post RO (ppb)	<7.5	<7.5
	Point of Use (ppb)	<7.5	<7.5
Experiment II	Pre-Spiking (ppb)	<7.5	<7.5
	Post 50 L bag (ppb)	<7.5	24.3
	Post RO (ppb)	<7.5	<7.5
	Point of Use (ppb)	<7.5	<7.5

Table 4: Total Organofluorine (TOF) via Combustion Ion Chromatography (CIC)

The Arium® Comfort I UV system consistently produces high-quality reagent water, making it well-suited for PFAS testing labs. The TOF data serves as a valuable complement to LCMS data, ensuring a more comprehensive analysis. Verified non-detects and the confirmed increase after a 50 L Bag in System 2 reinforce the system's reliability. Importantly, the impact of PFAS contribution is verified, dispelling concerns of hidden analytes driving total organic fluorine levels upward. This underscores the system's transparency and accuracy in PFAS analysis.

Return on Investment

The cost analysis provided over a one-year period, as shown in Figure 5 clearly highlights the advantages of transitioning to in-house treated water. With a daily demand of two litres, the investment in the treatment device starts to yield significant financial benefits in as little as four months, ultimately resulting in substantial cost savings within the course of a year. Furthermore, this approach aligns with sustainability goals by eliminating the need for glass bottles, reducing pollution associated with shipping, and eliminating the necessity for extensive storage space.

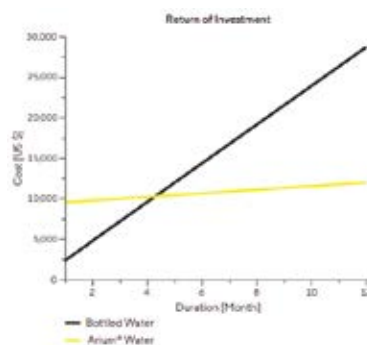


Figure 5: Comparison of Expenses Between In-House Produced Arium®-Water Type I (Ultrapure Water) and Bottled Water (LCMS Grade, With PFAS Content Below Detection Limit)

Conclusion

The Arium® Comfort I UV offers a comprehensive solution for the reduction of PFAS contaminants from tap water, reducing their concentrations to below limit of detection.

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Through a combination of activated carbon filtration, reverse osmosis treatment, and resin ion exchange, the system ensures the production of ultrapure water suitable for a wide range of applications. Additionally, the system's ability to treat highly contaminated solutions (~4 ppb) demonstrates its efficacy in managing even the most challenging PFAS contamination scenario, whether the system is equipped with PE tubing alone or both PE and PTFE tubing (Setup I and II). For analytical laboratory and applications demanding the highest water quality standards, the Arium® Comfort I UV provides a reliable and effective solution for PFAS reduction.

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